07 · Manufacturing: 100-Unit Run with PCBWay

Doc: LOT-HW-01-MFG · **Rev:** A (2026-11-14)

PCBWay does all four of the main manufacturing steps: PCB fabrication, PCBA (turnkey parts), CNC machining in stainless steel with polishing, and box build. Running all of them through one vendor makes the run simpler to manage.

Service	Link	What we order
PCB fabrication	https://www.pcbway.com/orderonline.aspx	4-layer, 0.6 mm (EVT) / 0.4 mm HDI (DVT), ENIG, 36 × 36 mm, panelised 3 × 3 with mouse bites
Flex PCB (optional)	https://www.pcbway.com/flexible-pcb-manufacturer.html	Camera/antenna/LRA flex, polyimide, 0.1 mm
PCB assembly	https://www.pcbway.com/pcb-assembly.html	Turnkey for commodity parts. Consigned: display, camera, cell
CNC machining	https://www.pcbway.com/rapid-prototyping/cnc-machining/	316L back tray and front bezel, #8 mirror polish, laser marking
Box build	https://www.pcbway.com/ (quote by email)	Final assembly, test and packaging (PVT)

1. BUILD PHASES AND QUANTITIES

Build	Qty	PCB	Enclosure	Purpose
EVT-0	3	Dev kits on breadboard	3D-printed (SLA)	Firmware and API bring-up
EVT-1	10	Custom PCB, module	5 CNC + 5 SLA	Electrical validation, RF in a steel box, Qi through the window
DVT	20	Chip-down if D-2 = 5 mm	CNC, final finish	Thermal, drop, battery life, cosmetics, pre-scan for FCC
PVT / run	**100 (+10 spares)**	Final	CNC + polish	Customer units

2. FILES PCBWAY NEEDS

- Gerbers (RS-274X) and NC drill, a stack-up note, and impedance control on the antenna feed (50 Ω)
- Centroid / pick-and-place (`.csv`) and assembly drawing (top and bottom)
- BOM in PCBWay format: export `docs/bom.csv` with columns `Designator, Qty, MPN, Manufacturer, Description, Package`
- STEP files for the enclosure plus 2D PDF drawings with tolerances (03 §7)
- Finish spec: "#8 mirror on the outer back face, Ra ≤ 0.05 µm; side walls #4 brushed vertical; no tool marks; protective film before shipping"

3. ESTIMATED RUN COST AT 100 UNITS (USD)

These are estimates for planning, not quotes.

Line	Estimate
Material (02-BOM, EVT variant)	10 425
PCB fab + PCBA labour, stencil, X-ray for QFN/LGA	1 800
CNC NRE / set-up, first-article samples	900
Test fixture (bed of nails, pogo, Tag-Connect, Qi pad)	600
Final assembly + test labour (~15 min/unit)	1 000
Freight (DDP) + battery dangerous-goods surcharge	600
Run subtotal	**≈ 15 300 (≈ $153 per unit)**
FCC (module path: Part 15B + modular grant reuse + SAR exempt check)	3 000 – 5 000
FCC (chip-down path: full Part 15C intentional radiator)	10 000 – 18 000

4. CERTIFICATION AND COMPLIANCE

- **FCC (USA):** the EVT/PVT module path reuses the Espressif modular grant, and the unintentional-radiator test (Part 15B) is still required. The label shows "Contains FCC ID: 2AC7Z-ESPS3MINI1" on the laser-marked back. Re-check the grant for the exact module variant.
- **Qi:** "Qi-compatible" wording only. "Qi Certified" (WPC logo) needs WPC membership and testing, which is not budgeted for 100 units.
- **Battery transport:** UN38.3 test summary and MSDS from the cell vendor, IATA PI 967 (cells inside equipment).
- **RoHS / REACH** declarations from PCBWay and the cell vendor. **Prop 65** check for the California market.
- **WEEE / battery** labels if selling to the EU (out of scope for the first 100).

5. FINAL FUNCTIONAL TEST (EACH UNIT)

1. Program and run the factory test (05 §7). The fixture logs pass/fail to `test-log.csv`, keyed by serial.
2. Qi: place on the reference pad. CHG_STAT must assert within 5 s, and charge current must be within ±15 %.
3. Wi-Fi: associate to the fixture AP. RSSI must be ≥ −65 dBm at 1 m, with the enclosure closed.
4. Camera: capture a target chart and check the JPEG hash range plus mean luminance.
5. Haptic: LRA back-EMF auto-calibration passes on the DRV2605L.
6. Cosmetic: inspect the back mirror under a D65 light booth, then fit the protective film.
7. Ship in "unpaired" state. The first press shows a pair code.